



Fig. 3: TCS3200 colour sensor module



Fig. 5: Xloader window

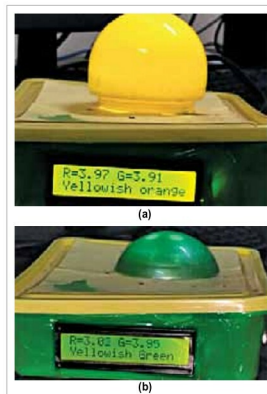


Fig. 4 (a and b): Detection of yellowish coloured objects

For instance, blue starts at 400nm, green ranges between 500 and 540nm, yellow at 575nm, and red at 640nm.

The complementary colours, including purple and magenta, are at the bottom of the chart. The white region shown at the centre is the approximate range of white colour, though some colours have coordinates falling within this white elliptical region also. The chart is precise, but the diagram shown here with demarcations between colours is only approximate.

Thus, the curved lines shown distinguishing, say green and yellowish green, are only approximate. In this

chart, the main colours are seventeen in number. Therefore the circuit and program in this article are meant for identifying up to twenty-two colours including complementary colours.

Circuit and working

The circuit diagram for colour identification is shown in Fig. 2. It is built around an Arduino Uno board (Board1), 16 × 2 LCD (LCD1), TCS3200 colour sensor module (CS), and a few discrete components. TCS3200 colour sensor module is shown in Fig. 3.

The circuit shown uses the frequency output. The external interrupt pin INT0 (pin 2) of Arduino is used to count the frequency values. We have selected 10kHz as the maximum range. Pins S2 and S3 select colours in the order of 00, 01, and 01. Only the so-called tristimulus values of red (R), green (G), and blue (B) are available. The details of the colour sensor module were already published in 'RGB Colour Detector Using TCS3200 Sensor Module' DIY article now available at

<https://www.electronicsforu.com/electronics-projects/rgb-color-detector-tcs3200-sensor-module>

The three frequency values are used to determine the chromaticity coordinates and thereby identify one among 22 classified colours from the CIE chromaticity chart using our search algorithm. The chromaticity coordinates (multiplied by ten) are indicated for every object along with

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno
CS	- TCS3200 colour sensor module
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 470-ohm
VR1	- 10-kilo-ohm potentiometer
Miscellaneous:	
LCD1	- 16×2 LCD
	- 8-pin female Berg connector for the sensor

the nearest colour name as per the colour demarcated CIE chart (Fig. 2). These R and G values are used for colour matching.

An LCD is interfaced to the Arduino board to show both the coordinates R and G and the colour name. The R and G are used for colour matching as two identical coloured objects will have similar values.

The circuit is also interfaced with a two-row character LCD display. Pin 3 of LCD is connected to a 10-kilo-ohm (VR1) potentiometer to adjust the voltage between 0 and 5V for varying the contrast of the characters shown on the LCD module.

The data comprises four wire connections to pins D4 through D7 of the LCD (11-14). Pin 15 is connected to +5V and used for the backlight of the LCD, and pin 16 is grounded via resistor R2.

Here, we have used a four-wire data connection rather than the eight-wire data connection. Therefore the program has to send the command and address values in two sequential write operations. Using four-wire connection saves pins on the microcontroller.

The yellowish objects being detected and displayed on the LCD, along with the values of the signals, are shown in Fig. 4.

Programming the chip

The program (colour_sense.bas) for the unit was developed on Oshon AVR

EFY Note

The source code of this project is available for free download at source.efymag.com